From agents to equations and back: dynamics, Lotka–Volterra approximations and harvest control in a spatial predator–prey ecosystem

Lotka project*

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Abstract

We study a spatial, stochastic agent-based model (ABM) of a vegetation–prey–predator food chain on a toroidal lattice as a test bed for core ideas of dynamic systems and control. The rules are minimal: predators kill adjacent prey and starve after N ticks without food, prey must graze K vegetation cells per tick and starve after S underfed ticks, and grazed cells regrow after R ticks. We compare the ABM with five Lotka–Volterra (LV) type descriptions whose rates are estimated directly from per-tick event counts, and add harvesting as a control input. Across a sweep of the predator starvation time, both species coexist for $10 \leq N \leq 40$ ticks; below this window predators die out, above it they overexploit their prey. Fitted LV equilibria reproduce the ABM's mean populations to within 2.7% on average, including the classical result that prey abundance is set by predator traits. Yet every fitted deterministic model predicts damped oscillations, whereas the ABM sustains cycles whose amplitude grows smoothly with N (prey coefficient of variation 0.04 to 0.51): the cycles are noise-sustained quasi-cycles, and their period is shorter than any fitted model predicts. Spatial structure lowers the effective predation rate 3.4-fold relative to the well-mixed limit, and the well-mixed mean-field of the same rules oscillates violently, so space is strongly stabilising. A quality-diversity search over 18 parameters exposes a robustness–amplitude trade-off and a specification-gaming failure mode (immortal, non-reproducing predators) that had to be excluded from the objective. Finally, open-loop quota harvesting collapses the ecosystem abruptly beyond a critical quota, constant-effort harvesting degrades gracefully, and proportional feedback on the predator population suppresses the predator–prey cycles at a given yield. All code and data are open.

1 Introduction

Predator–prey interactions are the textbook example of a nonlinear dynamic system. The Lotka–Volterra (LV) equations [1, 2] predict neutral cycles around an equilibrium whose prey level is set by predator parameters and vice versa. Their structural fragility is equally classic: adding prey self-limitation turns the centre into a stable focus, a saturating (type II) functional response can destabilise it into a limit cycle [3], and enrichment of the prey's resource pushes the system across that boundary, the *paradox of enrichment* [4]. Real and simulated populations add two ingredients the ODEs ignore: demographic noise, which can sustain oscillations in an otherwise damped system [5], and space, which lets local extinctions be recolonised and was shown experimentally to permit persistence [6]. Lattice predator–prey models have long been used to study these effects [7, 8], and individual-based teaching models such as the NetLogo wolf–sheep model [9] are widely used in classrooms.

This paper uses a deliberately simple lattice ABM for a different purpose: to make the vocabulary of systems and control—state, feedback loop, equilibrium, stability, delay, nonlinearity, model reduction, open- versus closed-loop control, and objective design for optimisation—*measurable* in a single system that students can run in a browser. We ask three questions.

1. **Dynamics.** Which qualitative regimes do the rules produce, and how do they depend on the predator starvation time N, the resource regrowth time R and system size?
2. **Reduction.** When does a two- or three-variable LV model, with rates estimated from the ABM's own event counts, describe the agent dynamics, and what does it miss?
3. **Control.** With harvesting as the input, how do open-loop (constant quota, constant effort) and closed-loop (proportional feedback) policies trade yield against stability?

We complement parameter sweeps with an evolutionary quality-diversity search over 18 parameters, which also illustrates a lesson every control engineer learns: optimisers exploit the objective you wrote, not the one you meant.

2 Model

The world is an $L \times L$ torus ($L = 200$ unless stated). Each cell has a *vegetation layer* (grown or regrowing; a fraction of cells may be barren) and an *animal layer* holding at most one animal: empty, prey or predator. Animals carry a hunger counter and an age. One tick executes, in order:

1. **Regrowth.** A grazed fertile cell regrows exactly R ticks later.
2. **Movement.** Prey, then predators, move to a free Moore neighbour. With probability 0.7 the move is directed: prey maximise local vegetation minus $5\times$ local predator density (sensing radius 3); predators maximise local prey density (radius 4). Otherwise the move is random.

3. **Predation.** Each predator kills prey in its Moore neighbourhood, at most one per tick (satiation).
4. **Grazing.** Each prey eats up to K grown cells in its 3×3 neighbourhood (own cell first); it is *fed* if it ate K ($K = 1$ here).
5. **Hunger and ageing.** Unfed prey increment hunger and die at S ($= 6$); predators without a kill increment hunger and die at N ($= 12$ by default). Optional maximum ages.
6. **Reproduction.** A fed prey reproduces with probability 0.08, a predator that killed with probability 0.3, into a random free neighbour.
7. **Harvest (control input).** u animals of the harvested species are removed at random: constant effort $u = \mathrm{Bin}(n, E)$, constant quota $u = \min(Q, n)$, or proportional feedback $u = \mathrm{clip}\big(\lfloor K_p(n - r)\rceil,\, 0,\, n\big)$ with set point r, switched on at a given tick.

Initial densities are 60% vegetation, 6% prey and 1% predators. Conflicts (two agents targeting one cell) are resolved in random order; runs are reproducible from a seed. The model is vectorised in NumPy ($\approx$3 ms per tick at 200^2). We record per tick the counts x (prey), y (predators), v (vegetation) and every event: births, kills, starvation, age deaths, cells grazed and regrown, and harvest. The event identity $x_{t+1} = x_t + \text{births} - \text{kills} - \text{starved} - \text{aged} - \text{harvested}$ is checked by the test suite.

3 Methods

Metrics. Statistics use the second half of each run (after the transient): mean, coefficient of variation (CV, oscillation amplitude), dominant period (first autocorrelation maximum after the first zero crossing, accepted if > 0.1), predator lag (argmax of the prey–predator cross-correlation), and outcome (coexistence, predator extinction, total extinction). Periods are reported only for coexisting runs with prey CV ≥ 0.05. Unless stated, each condition uses 5–10 seeds and 3000 ticks.

LV family and event-based fitting. With x, y, v as densities per cell, we consider classic LV ($\dot{x} = ax - bxy$, $\dot{y} = cbxy - gy$), logistic LV ($ax \to rx(1 - x/K)$), Rosenzweig–MacArthur (RM; $bxy \to axy/(1 + ahx)$), and a three-level chain $\dot{v} = \rho(V - v) - evx$, $\dot{x} = \varepsilon evx - \mu x - bxy$, $\dot{y} = cbxy - gy$. Instead of fitting trajectories, which drift out of phase for noisy cycles, each rate is a regression of the matching event count on the pre-tick state, e.g. kills $\approx b\,x\,y$, predator births $\approx$ c·kills, predator deaths $\approx g\,y$ (RM via Gauss–Newton). Unit tests recover known rates exactly from synthetic histories. Models are integrated (RK4) from the ABM's initial state, and equilibria are classified by the eigenvalues of the numerical Jacobian, giving a predicted period $2\pi/\mathrm{Im}\,\lambda$ and damping time $-1/\mathrm{Re}\,\lambda$. We also derive, without fitting, a *mean-field map* that applies the exact rules to a well-mixed population (binomial neighbourhoods, hunger-class structure, delayed regrowth).

Experiments. E1: seven regimes (Table 1). E2: $N \in [2, 60]$, 17 values $\times$ 6 seeds. E3: $R \in [4, 60]$ for $N = 12$ and 30. E4: $L \in [40, 280]$ in two regimes, 10 seeds. E5: quality-diversity search. E6: harvesting (effort, quota, feedback, open vs closed loop). E7: full model comparison in two regimes. E8: shocks (-50% predators) at $t = 1500$, 16 seeds per condition.

Optimisation. A genome is a vector of 18 parameters in the unit hypercube (vegetation, prey and predator traits, initial densities). Each genome is simulated on 3 training seeds (common random numbers), 100^2 cells, 1500 ticks. Fitness for *robust coexistence* is mean survival fraction $+\,0.5\times$resilience (minimum/mean population after the transient) $-0.25\times$ the std of survival. MAP-Elites [10] keeps the fittest genome in each cell of a 12×12 behaviour grid (prey CV $\times$ prey period); offspring arise by BLX-α crossover and Gaussian mutation of archive members. The top genomes are re-validated on 10 hold-out seeds. A target-free objective is also offered for harvest yield (Sec. 4.10).

4 Results

4.1 Regimes

The rules produce a rich set of behaviours (Fig. 2, Table 1). With default parameters the populations settle to a noisy equilibrium of $\approx$ 2132 prey and 510 predators (prey CV 0.06). Raising N from 12 to 30 produces sustained cycles (CV 0.36, period $\approx$ 142 ticks) in which predators lag prey by $\approx$ 44 ticks, close to the quarter period expected for LV dynamics. Spatially, the cycles appear as travelling fronts of prey invading vegetation and predators following them (Fig. 1). Cutting the predation loop ($N = 3$) leaves prey limited only by vegetation at ≈ 3508; overly efficient predators eat all prey and then starve.

4.2 Predator starvation time sets the regime

Figure 3 sweeps N. Coexistence is certain for $10 \leq N \leq 40$. For $N \leq 8$ predators cannot bridge the time between kills and go extinct, after which prey sit at the vegetation-limited carrying capacity. At $N = 50$ only 50% of seeds coexist: long-lived predators overshoot and collapse their food supply (overexploitation).

Inside the window, mean prey fall from 2132 at $N = 12$ to 903 at $N = 40$ while predators saturate at ≈ 718. This is the LV signature: the prey equilibrium $x^* = g/(cb)$ depends on the predator death rate g, which falls with N, whereas y^* is set by prey production. The equilibria of the fitted chain model reproduce the ABM means with a mean error of 2.7% (maximum 10%; Fig. 3b).

4.3 Damped models, sustained cycles

The oscillation amplitude grows smoothly with N, from a prey CV of 0.04 to 0.51 (Fig. 3c); there is no sharp onset. Every fitted logistic, RM and chain model in the window has a *stable focus*, i.e. predicts that oscillations decay. Table 2 and Fig. 4 show the cycling regime in detail: all fitted models get the means right, the deterministic ones relax to equilibrium, and classic LV oscillates only because it is neutrally stable, with an amplitude (CV 0.84) that depends on initial conditions rather than dynamics. The ABM's cycles are therefore

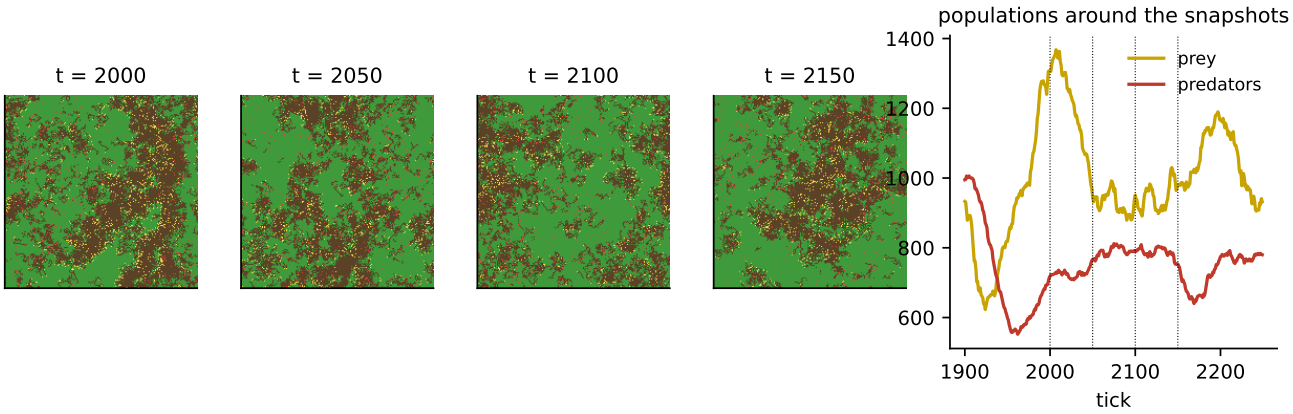


Figure 1: Cycling regime ($N = 30$). Grid snapshots (green vegetation, brown regrowing soil, yellow prey, red predators) during one cycle and the corresponding population curves.

Table 1: Regimes (E1): 200^2 cells, 3000 ticks, 6 seeds. Means and CV over the second half; period only where prey CV≥ 0.1 (visible oscillations). "Prey only" has no predators by design; for total extinction the extinction tick is given.

Regime	Change from defaults	Typical outcome	Prey	Predators	Prey CV	Period
Stable coexistence	defaults (N=12, R=20)	coexist	2132	510	0.06	–
Sustained cycles	N=30	coexist	1036	729	0.36	142
Enrichment	N=30, R=10	coexist	3311	671	0.60	142
Predator starvation	N=3	predators extinct	3508	0	0.01	–
Overexploitation	kill radius 2, 8 kills, N=40	all extinct (t≈62)	–	–	–	–
Prey only	no predators, fast breeding	predators extinct	3579	0	0.06	–
Vegetation scarcity	30% fertile, R=30	predators extinct	426	0	0.06	–

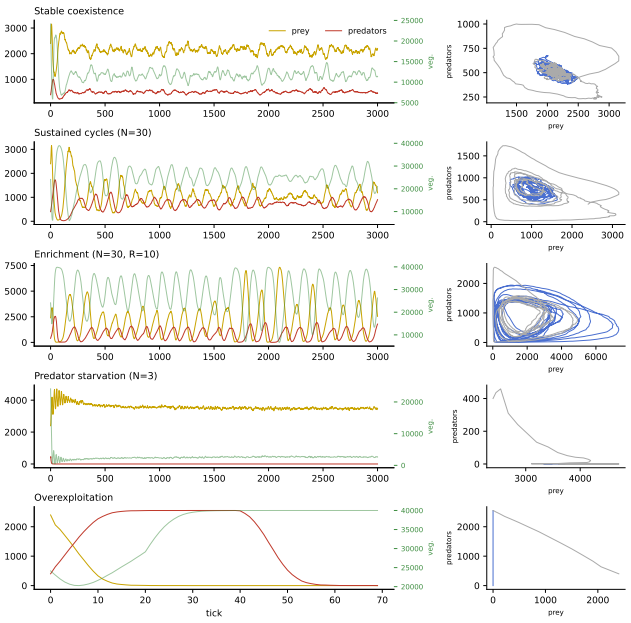


Figure 2: Time series (left; vegetation on the right axis) and phase portraits (right; grey early, blue late) for five regimes.

best understood as *noise-sustained quasi-cycles* [5]: demographic noise continually excites a weakly damped mode. Increasing N weakens predator mortality, which pushes the leading eigenvalue towards the imaginary axis (fitted RM damping time 49 ticks at $N = 12$ vs 274 at $N = 30$), so the same noise produces larger oscillations.

The ABM cycles are also faster: periods of 112–162 ticks for $N \geq 15$ against 164–321 (chain) and 231–329 (RM) predicted from the eigenvalues (Fig. 3d). ODEs replace the fixed starvation delay by an exponential death rate g; a constant g explains only $R^2 = 0.14$ of the per-tick predator deaths. A delayed death process shortens the feedback loop's response compared with the exponential approximation of the same mean lifetime.

Table 2: Cycling regime ($N = 30$, seed 42, 3000 ticks): ABM vs models fitted to its events. Stability from the Jacobian at equilibrium; period from the eigenvalues where available.

	Equil.	Prey	Pred.	CV	Period
ABM (spatial)	–	975	756	0.24	138
Classic LV	centre	1004	612	0.84	247
Logistic LV	stable focus	979	781	0.00	238
Rosenzw.–MacArthur	stable focus	918	751	0.00	254
Vegetation chain	stable focus	979	747	0.00	201
Mean-field	–	2476	194	0.96	398

4.4 Space is stabilising

In a well-mixed population a predator with eight neighbour cells meets prey at rate $\approx 8x$, so $b \approx 8$. The rate fitted from kills in the ABM is 2.2–2.5 across the coexistence window, a 3.4-fold reduction (Fig. 5): predators deplete their surroundings and must move to find new prey, while prey that escape form clusters behind the front. The mean-field map of the same rules, which assumes mixing, is wildly oscillatory (prey CV ≥ 0.91 in the whole window, against 0.04–0.51 in the ABM) and keeps predators near extinction between

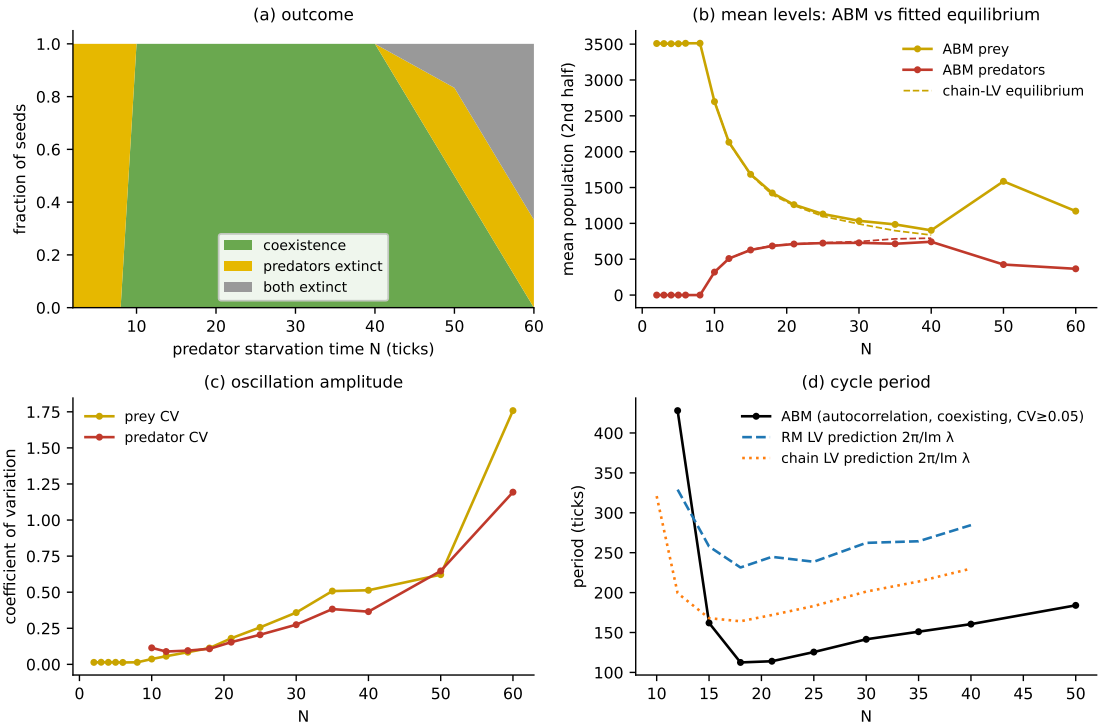


Figure 3: Sweep of predator starvation time N (E2; 6 seeds per point, 3000 ticks, 200^2). (a) Outcomes. (b) Mean populations vs equilibria of the fitted chain model (dashed). (c) Oscillation amplitude. (d) Cycle period of the ABM vs the eigenvalue predictions of the fitted RM and chain models.

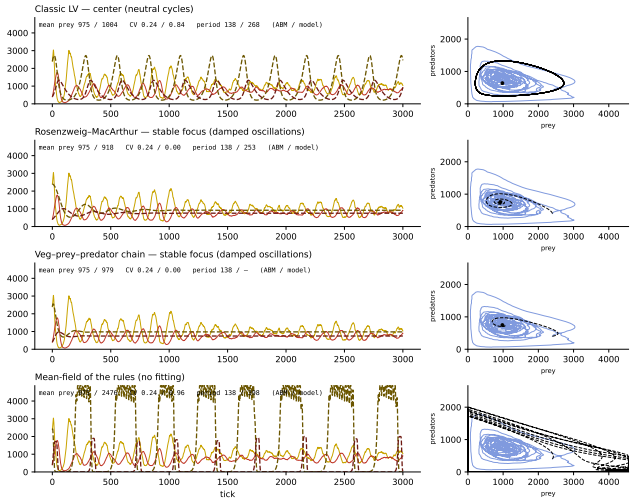


Figure 4: ABM (solid) vs models (dashed) in the cycling regime, with phase portraits. Numbers: ABM / model.

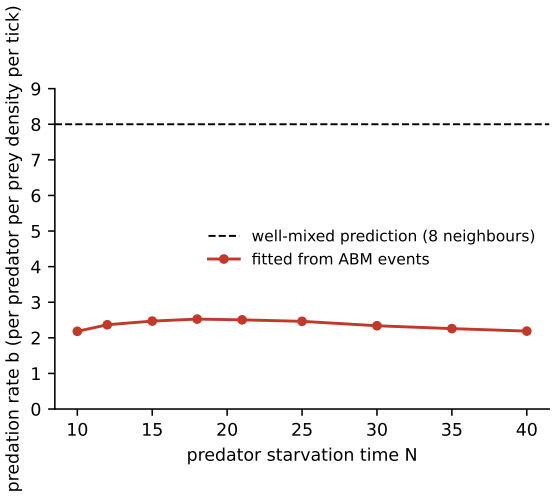


Figure 5: Predation rate b fitted from ABM kill events vs the well-mixed prediction.

outbreaks (mean 208 vs 510 in the stable regime). Local depletion thus acts as a distributed negative feedback that the well-mixed description lacks [8].

4.5 Enrichment: coexistence is bounded on both sides

Shortening the regrowth time R enriches the prey's resource (Fig. 6). For $N = 12$ the prey CV rises from 0.09 at $R = 60$ (where predators are already extinct) to 0.30 at $R = 4$, and peak prey reach $\approx$ 10219. In the cycling regime ($N = 30$) the effect is stronger and lethal: amplitude grows to CV $\approx$ 0.56, and coexistence drops to as low as 0% of seeds at $R = 8$ (boom–bust cycles drive predators, and sometimes prey, to extinction). This is the paradox of enrichment [4] reproduced by agents, with a qualitative difference from the RM model: here destabilisation is gradual and acts by amplifying stochastic troughs until one hits zero, rather than through a sharp Hopf bifurcation. On the poor side, slow regrowth ($R \geq 28$ for $N = 12$, $R = 60$ for $N = 30$) keeps prey too sparse to sustain predators (bottom-up control). Coexistence thus occupies an intermediate band of resource productivity whose width shrinks as predators become more persistent. Richer resources also raise the mean prey level, which a chain model with a fixed $x^* = g/(cb)$ cannot express; the shift reflects nonlinear averaging over larger

cycles.

4.6 System size and demographic noise

In the stable regime the prey CV scales as $(\text{cells})^{-0.48}$ (Fig. 7a), the $-\frac{1}{2}$ law of independent demographic fluctuations: the deterministic equilibrium is approached as the system grows, and the noisy wobble is a finite-size effect. The cycling regime decays more slowly (exponent -0.41 over the sizes where it persists): its amplitude is set by the resonant amplification of noise near a weakly damped mode as well as by system size. Persistence depends strongly on size (Fig. 7b): the stable regime survives in 80% of 40^2 worlds and in all worlds from 60^2 cells, whereas the cycling regime never survives at 40^2 and needs 140^2 cells for certain persistence. Large oscillations combined with small populations make extinction by a random trough likely, a stochastic analogue of an unstable equilibrium. (The optimisation of Sec. 4.7 used 100^2 cells, where even the cycling preset persists in only 80% of runs; its fitness therefore rewards robustness to this finite-size risk.)

4.7 Optimisation: the space of behaviours

The MAP-Elites search (25 generations, population 48, 3744 simulations, with finite lifespans of 150/300 ticks) filled 81 of 144 behaviour cells (Fig. 8). Coexistence is possible from almost static populations to violent cycles in which prey vary many-fold (CV up to ≈ 1.4; e.g. cell (10,6): CV 1.36, period 230, prey from 6 to 4070). Fitness falls monotonically with amplitude, from 1.48 for CV< 0.125 to 1.01 for CV≥ 1.25: a *robustness–amplitude trade-off*, since deep troughs leave little margin before extinction. The validated best genome (fitness 1.476 on 10 hold-out seeds) is a dense, nearly static community (4254 prey, 1744 predators, CV 0.014) with satiation raised to 4 kills per tick.

Specification gaming. A first search without lifespans and without a reproduction requirement converged on a degenerate optimum: predators with zero reproduction probability and long starvation time, i.e. a few dozen immortal predators living among prey that filled 94% of the grid. It "coexisted" perfectly and contained no predator–prey dynamics at all, a textbook case of an optimiser exploiting an incomplete objective [11]. We excluded it by (i) imposing finite lifespans during optimisation and (ii) counting a run as coexistence only if both species are still being born in the second half. In the corrected archive the smallest evolved predator reproduction probability is 0.029.

4.8 Harvest control

We now treat harvesting as the control input u (Fig. 9) acting on the stable regime (open loop) and on the cycling regime (closed loop).

Constant effort and the Volterra principle. Removing a fixed fraction E per tick behaves as LV theory predicts (Fig. 10). Harvesting prey does not lower the prey level: it *lowers predators* (from 506 to 387 at $E = 0.01$) while prey stay at or slightly above their unharvested level (2209 vs 2140), because $x^* = g/(cb)$ does not contain the prey's growth rate. Harvesting both species at $E = 0.010$ raises prey to 2494 and cuts predators to 293: Volterra's explanation for the rise of predatory fish when Adriatic fishing stopped during the First World War [2]. The largest prey yield compatible with coexistence in every seed is 47.4 prey/tick at $E = 0.020$; from $E = 0.030$ predators are lost, after which the prey-only system follows a Schaefer curve [12] with maximum yield 106 at $E = 0.050$. The predator yield peaks at 6.4 per tick ($E = 0.020$), and predators vanish from $E = 0.040$.

Constant quota: abrupt collapse. A quota Q ignores the state. Up to $Q = 60$ prey/tick (switched on at $t = 500$) the system absorbs it, yielding exactly Q with predators reduced to 171 (Fig. 11). One step further, at $Q = 70$, only 60% of seeds survive, and at $Q = 80$ both species are extinct in every seed. The quota sustains a higher yield than constant effort (60 vs 47.4) precisely because it keeps taking when the stock declines, which is also why its failure is total rather than gradual: the harvested fraction Q/x grows as x falls, a destabilising feedback [13]. Effort harvesting has the opposite, stabilising feedback built in; past its limit it loses the predators but the prey persist. For predators the safe quota is 4 per tick and $Q = 6$ is already lethal.

Proportional feedback suppresses cycles. In the cycling regime we switch on $u = K_p \max(y - r, 0)$ on predators at $t = 1500$ (Fig. 12). Even a small gain is effective: with $r = 400$ the predator CV after switch-on falls from 0.30 to 0.01 at $K_p = 0.2$ and the prey CV from 0.40 to 0.03, with coexistence in all seeds. The controller caps predator peaks, removing the delayed overshoot that drives the cycle; prey, though not measured, are stabilised through the coupling. Three textbook features of feedback [14] appear. First, a *steady-state error*: the predator mean is 738 at $K_p = 0.02$, 480 at $K_p = 0.2$ and 400 at $K_p = 1$ against a target of 400. Second, *actuator saturation*: harvesting can only remove animals, so below the target the loop is open. Third, a gain–effort trade-off: the harvest the controller must take grows with gain and saturates around 19.9 predators/tick.

Controlling the prey instead reveals a hidden-state hazard: holding prey at $r = 800$ with $K_p \geq 0.5$ drives predators extinct in every seed, because the target lies below the prey density predators need to break even. A set point on one state variable can destroy an unmeasured one—the ecological analogue of unstable internal dynamics behind a well-regulated output.

Open versus closed loop at equal yield. Comparing constant predator effort with feedback ($K_p = 0.2$, varying r; E6d, both from $t = 0$) at matched predator yields of 9.0–14.4 per tick, the feedback policy leaves a prey CV that is on average 0.20 times that of the open-loop policy (Fig. 13). Constant effort also damps cycles, since it adds predator mortality, but never brings the prey CV below 0.12 in the tested range, whereas feedback reaches 0.02.

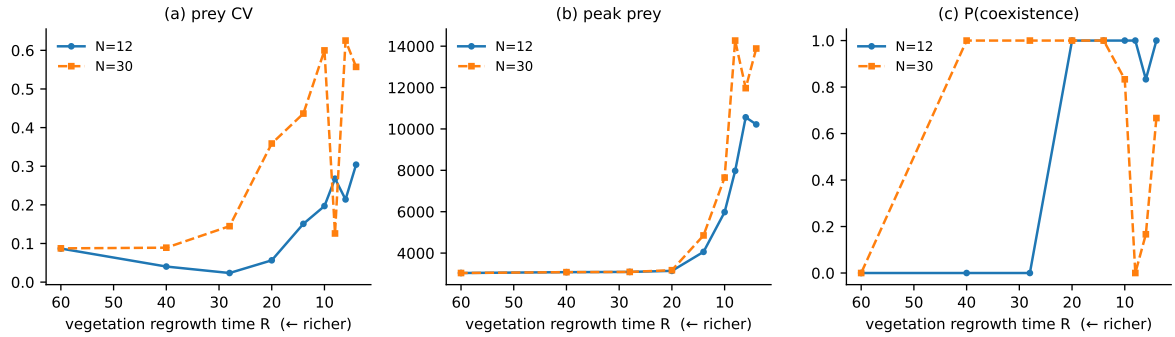


Figure 6: Enrichment (E3; 6 seeds per point): vegetation regrowth time R (richer to the right) for a stable ($N = 12$) and a cycling ($N = 30$) predator.

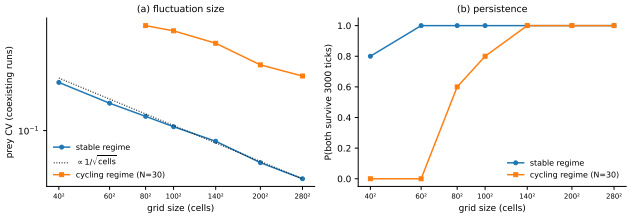


Figure 7: System size (E4; 10 seeds). (a) Prey CV in coexisting runs; dotted: $\propto 1/\sqrt{\text{cells}}$. (b) Probability that both species survive 3000 ticks.

4.9 Stability probe

Removing half of the predators at $t = 1500$ (Fig. 14) produces, in the stable regime, a prey outbreak peaking ≈ 608 above control after 20 ticks, and the deviation envelope decays by a factor e in ≈ 39 ticks (16 seeds). The fitted chain model predicts a damping time of 41 ticks: the impulse response measured on agents and the eigenvalue computed from the reduced model agree. In the cycling regime the same shock does not simply decay: it resets the phase of the cycle in all seeds at once, and the mean deviation keeps oscillating for the full 900 observed ticks (Fig. 14b), the signature of a very weakly damped oscillatory mode—consistent with the noise-sustained cycles of Sec. 4.3.

4.10 Optimising the harvest policy

Finally we let a genetic algorithm design the prey-harvest policy: the genome holds the controller type (effort, quota or feedback) and its parameters, and fitness is the mean prey yield per 1000 cells in the second half of a 1500-tick run on a 100^2 grid (3 training seeds, population 32, 20 generations, 1896 simulations). No stability or coexistence term was included. The search converged on constant effort with $E = 0.028$, a training fitness of 1.79 prey per tick per 1000 cells. Among policies that kept both species alive on all training seeds, the best effort, quota and feedback designs reached 1.79, 0.50 and 0.05, respectively. Feedback was abandoned early: only 11 of 632 genomes used it, and 9 of those had set points above the prey level of a 100^2 world, so the controller was idle. The GA's verdict on feedback is therefore uninformative, not negative.

Re-validated on 10 fresh seeds, the same genome scored only 0.71: predators survived in 50% of seeds (mean 24 predators). The evolved effort sits just below the point where predators are lost in E6a, the edge of the coexistence region, where the outcome depends on chance. This is the expected behaviour of an optimiser given a yield-only objective with few seeds: it climbs to the boundary of the feasible region and *overfits* the training noise. Robust management requires putting the constraint (coexistence, variability) into the objective and evaluating it over many stochastic realisations—the same lesson as the specification gaming of Sec. 4.7, now with an economic objective.

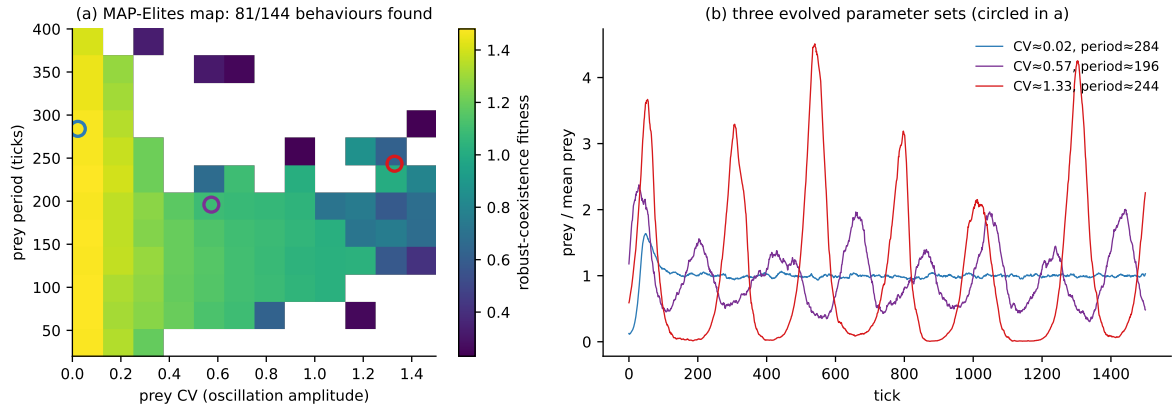


Figure 8: (a) MAP-Elites archive over prey CV × period (100^2 cells, 1500 ticks, 3 seeds per genome); colour is robust-coexistence fitness. (b) Prey time series (normalised) of three evolved genomes circled in (a).

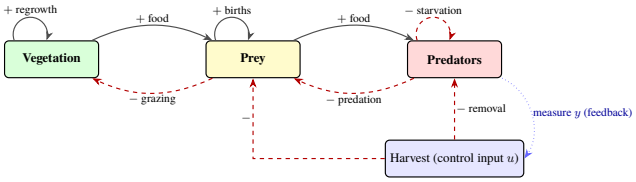


Figure 9: Causal-loop view of the ecosystem with harvest as control input. Solid: reinforcing (+); dashed: opposing (−); dotted: measurement used by feedback control.

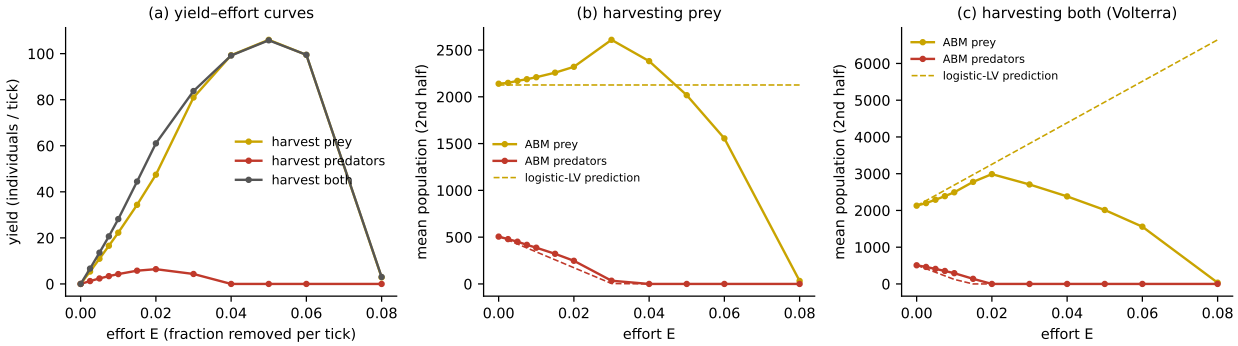


Figure 10: Constant-effort harvesting in the stable regime (E6a; 5 seeds, 3000 ticks). (a) Yield vs effort. (b,c) Mean populations vs effort when harvesting prey and both species; dashed: logistic-LV equilibrium calibrated without harvest, valid only while predators persist.

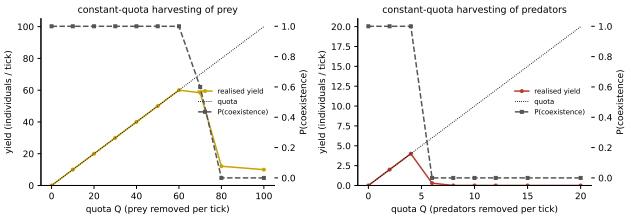


Figure 11: Constant-quota harvesting (E6b; 5 seeds; quota from $t = 500$): realised yield and probability of coexistence.

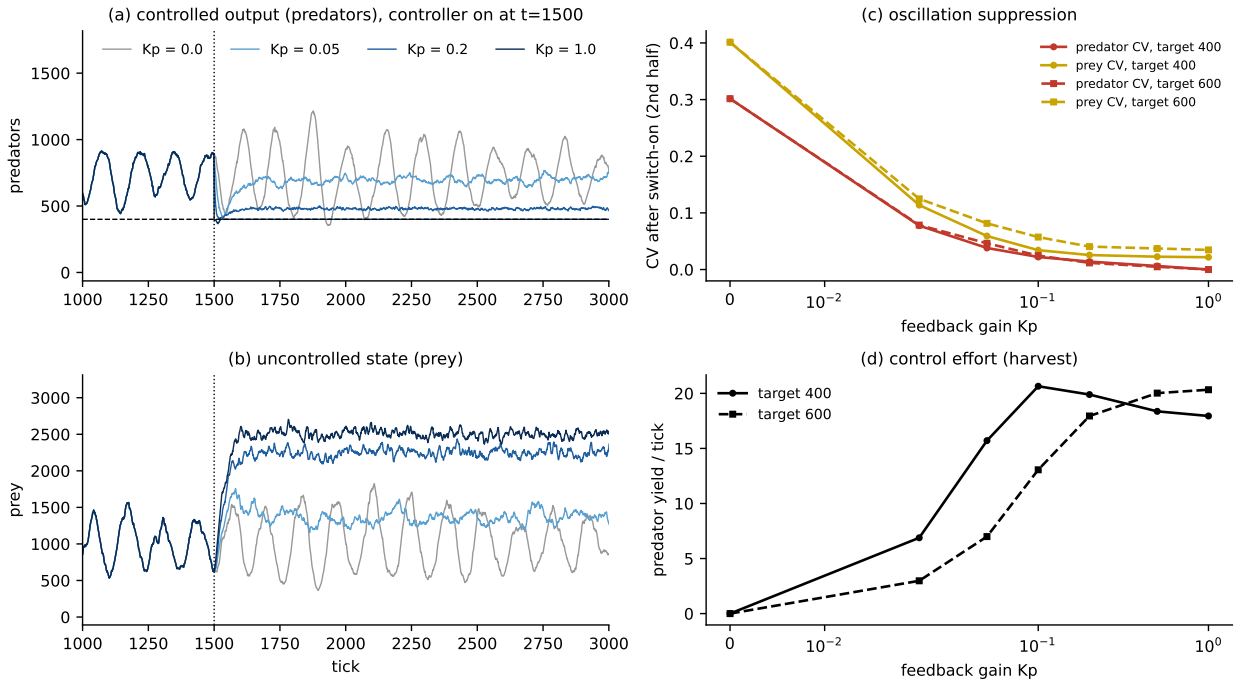


Figure 12: Proportional feedback on predators in the cycling regime (on at $t = 1500$; E6c, 5 seeds, 4000 ticks): controlled output, prey, post-switch CV vs gain, and harvest taken.

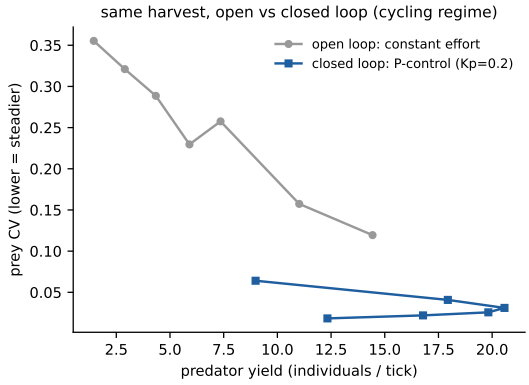


Figure 13: Prey variability vs predator yield for open-loop constant effort and closed-loop proportional control, cycling regime (E6d; 5 seeds).

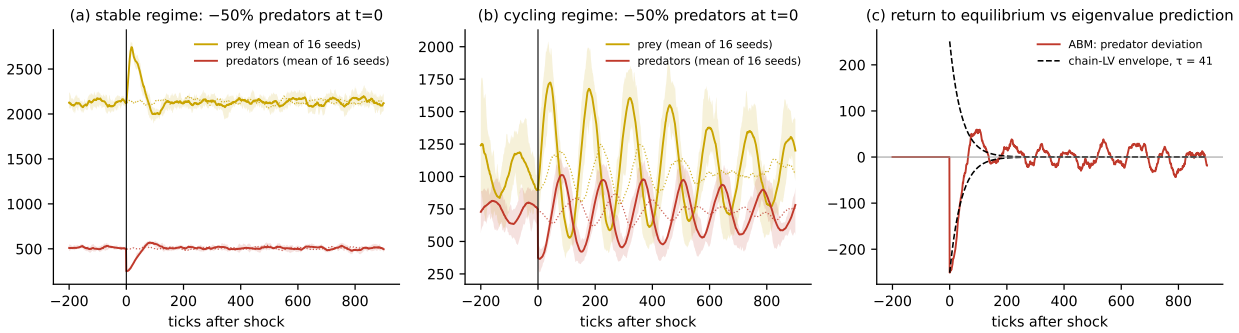


Figure 14: Mean response (16 seeds) to removing 50% of predators or doubling prey at $t = 1500$, relative to unshocked runs with the same seeds.

5 Discussion

What the reduction gets right and wrong. Rates estimated from event counts produce LV-type models that reproduce the ABM's mean populations, and with them the counter-intuitive structural predictions of LV theory: prey abundance is controlled by predator traits, harvesting prey lowers predators, harvesting both raises prey, and a pulse perturbation in the stable regime decays on the time scale given by the Jacobian's eigenvalues. Equilibria and their local stability are therefore robust, transferable properties. The reduction fails in three places, each of which corresponds to a modelling assumption that students can name: (i) oscillation *amplitude*, because deterministic models discard demographic noise, which sustains quasi-cycles around a stable focus; (ii) oscillation *period*, because a fixed starvation delay is replaced by exponential mortality; and (iii) *well-mixing*, which inflates the encounter rate several-fold and turns a stable spatial system into a violently oscillating mean field.

Space and noise pull in opposite directions. Local depletion lowers the effective predation rate and stabilises; demographic noise excites the weakly damped mode and destabilises by deepening troughs. Which wins depends on system size (Fig. 7) and on how close the mode is to the imaginary axis, which is set by predator persistence N and resource productivity R. The coexistence region in (N, R) is accordingly bounded on all sides: too little predator persistence or too little food leads to predator extinction; too much persistence or too much food leads to boom–bust cycles whose troughs are fatal in a finite world.

Control lessons. The harvesting experiments reproduce in a few minutes of simulation several lessons usually taught with linear models: open-loop policies are only as good as the model behind them and can fail catastrophically (quota); proportional feedback rejects disturbances and damps oscillations but leaves a steady-state error; actuator constraints (one-sided harvesting) open the loop; and regulating one output can destabilise an unobserved internal state. Feedback achieved a several-fold steadier ecosystem than constant effort at the same yield, a quantitative argument for state-dependent management rules [13].

Optimisation as an experimental tool. Quality-diversity search is useful not only to find parameters but to map the trade-offs a system permits; here, between amplitude and robustness. Its first result was also a warning: the optimiser satisfied the letter of "both species survive" with an ecosystem in which nothing happened. Writing objectives that encode the intended dynamics (turnover, finite lifespans) is part of the modelling.

Limitations. The rules are deliberately minimal: no sex, age structure, energy budgets or evolution of traits, a single vegetation type and fixed sensing radii. Seeds per condition (5–10) resolve the large effects reported here but not fine structure, e.g. the exact location of the coexistence boundaries. Periods are estimated from the autocorrelation of 1500-tick windows and are uncertain by roughly 10%. The comparison models are fitted to single runs.

6 Conclusion

A lattice with three rules per species reproduces the core phenomena of predator–prey theory—coupled oscillations with predators lagging prey, top-down control of prey abundance, the paradox of enrichment, overexploitation and Volterra's harvesting principle—and exposes where the classical equations fall short: noise-sustained amplitude, delay-shortened periods and spatial stabilisation. Treating harvest as an input turns the same system into a control laboratory in which open- and closed-loop strategies, steady-state error, saturation and hidden-state instability can be measured directly. All experiments are reproducible with `python -m experiments.exp all` and the companion guide turns them into exercises for first-year systems and control students.

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